Shuyi Ren

Postdoctoral Researcher in Decentralized Optimization and Wireless communication

Linköping, Sweden | shuyi.ren@liu.se | +46 70 455 38 05 | linkedin.com/in/suer2020

Academic Profile

Postdoctoral researcher at Linköping University working on decentralized optimization and consensus. Ph.D. in Computer and Information Engineering with research on optimization methods for wireless communications and intelligent reflecting surfaces, supported by undergraduate training in statistics and mathematical sciences. Research interests include optimization theory and algorithms, distributed learning, consensus, wireless communications, machine learning, and mathematical statistics.

Academic Appointment

Linköping University, Linköping, Sweden Feb. 2025 - Present

Postdoctoral Researcher, Division of Communication Systems, Department of Electrical Engineering (ISY)

Supervisor: Prof. Erik G. Larsson

- Conduct theoretical research on decentralized optimization and consensus for networked systems.
- Develop and analyze optimization algorithms that account for communication constraints and network structure.
- Apply the resulting theory and algorithms to concrete problems in wireless communications and machine learning.

Education

The Chinese University of Hong Kong, Shenzhen, China Awarded Nov. 2024

Ph.D. in Computer and Information Engineering

Supervisor: Prof. Zhi-Quan (Tom) Luo; Co-supervisor: Prof. Kaiming Shen

Dissertation: *Optimal Configuration of Intelligent Reconfigurable Surface* Defended Apr. 2024

Selected contributions

- Developed linear-time methods for discrete intelligent reflecting surface phase configuration, including optimal binary beamforming and an optimal algorithm for K-ary phase beamforming.
- Developed a scalable blind beamforming strategy that optimizes phase shifts without channel-state information and uses a polynomial number of samples to achieve quadratic signal-to-noise-ratio scaling.
- Formulated a joint optimization framework for aerial intelligent reflecting surfaces in heterogeneous networks to improve throughput and fairness.

Tongji University, Shanghai, China Sep. 2014 - Jun. 2018

B.Sc. in Mathematical Sciences, Statistics Track *Honors Class*

- First Honor Scholarship of Shanghai; Outstanding Student Leader.
- Foundation in mathematical analysis, probability, statistics, sampling, regression, and numerical computing.

Research Areas

Optimization	Decentralized optimization, distributed algorithms, consensus, resource-constrained optimization
Statistical methods	Mathematical statistics, stochastic processes, statistical signal processing
Networked systems	Wireless communications, intelligent reflecting surfaces, distributed learning

Journal Publications

1. Y. Zhang, K. Shen, **S. Ren**, X. Li, X. Chen, and Z.-Q. Luo, “Configuring intelligent reflecting surface with performance guarantees: Optimal beamforming,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 5, pp. 967-979, 2022.
2. **S. Ren**, K. Shen, Y. Zhang, X. Li, X. Chen, and Z.-Q. Luo, “Configuring intelligent reflecting surface with performance guarantees: Blind beamforming,” *IEEE Transactions on Wireless Communications*, vol. 22, no. 5, pp. 3355-3370, 2023.
3. **S. Ren**, K. Shen, X. Li, X. Chen, and Z.-Q. Luo, “A linear time algorithm for the optimal discrete IRS beamforming,” *IEEE Wireless Communications Letters*, vol. 12, no. 3, pp. 496-500, 2023.

Conference Papers

1. Y. Zhang, **S. Ren**, K. Shen, and Z.-Q. Luo, “Optimal discrete beamforming for intelligent reflecting surface,” *IEEE Global Communications Conference (GLOBECOM)*, pp. 1-6, 2021.
2. W. Wang, W. Lai, **S. Ren**, et al., “Adaptive beamforming for non-line-of-sight IRS-assisted communications without CSI,” *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2023.
3. **S. Ren**, B. Huang, X. Li, and K. Shen, “Aerial-IRS placement and load balancing in downlink multi-cell network,” *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2024.

4. **S. Ren**, Z. Chen, and E. G. Larsson, "Topology design for distributed consensus with relay-assisted communication," *IEEE International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, accepted, 2026.
5. **S. Ren**, N. Michelusi, and E. G. Larsson, "Budget-constrained multi-consensus decentralized gradient descent," *Allerton Conference on Communication, Control, and Computing*, accepted, 2026.

Manuscript Under Review

- C. He, **S. Ren**, J. Mao, and E. G. Larsson, "DeMuon: A decentralized Muon for matrix optimization over graphs," arXiv: 2510.01377; submitted to *Transactions on Machine Learning Research*.

Intellectual Property

- U.S. Patent Application US20240072848A1, "Method and system for controlling phase shifts of intelligent reflecting surface."

Selected Research Contributions

- **Decentralized optimization and consensus:** Studies optimization algorithms over networks, with particular attention to convergence, communication cost, and topology-dependent performance.
- **Discrete and blind beamforming:** Develops provably efficient algorithms and statistical methods for intelligent reflecting surface configuration under limited information.
- **Wireless network optimization:** Formulates resource-allocation and load-balancing models for heterogeneous communication systems.

Teaching Experience

Calculus, MAT1001 - Exemplary Leading Teaching Assistant	CUHK-Shenzhen
Linear Algebra, MAT2040 - Teaching Assistant	CUHK-Shenzhen
Time Series, SAT4003 - Leading Teaching Assistant	CUHK-Shenzhen
Multidimensional Statistics, SAT4002 - Leading Teaching Assistant	CUHK-Shenzhen

Academic Service

Reviewer: IEEE Transactions on Wireless Communications; IEEE Transactions on Communications; IEEE Wireless Communications Letters; IEEE Wireless Communications and Networking Conference; IEEE Global Communications Conference.

Selected Presentations

Optimization in Communication	Invited talk, LTH, Lund University
Aerial-IRS Placement and Load Balancing in Downlink Multi-Cell Network	Poster, ICASSP
Blind Beamforming for Intelligent Reconfigurable Surface	Poster, Graduate Research Conference
Beamforming for Intelligent Reconfigurable Surface	Internal presentation
User Association for Load Balancing in Heterogeneous Cellular Networks	Internal presentation

Honors and Awards

- Exemplary Teaching Assistant, The Chinese University of Hong Kong, Shenzhen.
- Poster Award, Graduate Research Conference, The Chinese University of Hong Kong, Shenzhen.
- First Honor Scholarship of Shanghai.

Technical Skills

Technical tools	MATLAB, Python, R, C++, SPSS, SAS, SQL Server
Languages	Mandarin Chinese, English, basic Swedish
Experimental work	Development of a practical testing platform for intelligent reflecting surfaces

References

Prof. Erik G. Larsson
Professor and Head of Division
Linköping University
erik.g.larsson@liu.se

Prof. Zhi-Quan (Tom) Luo
Presidential Chair Professor
The Chinese University of Hong Kong, Shenzhen
luozq@cuhk.edu.cn

Prof. Kaiming Shen
Assistant Professor, The Chinese University of Hong Kong, Shenzhen
shenkaiming@cuhk.edu.cn